

SAT Hard Question Set 15

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Phase 2 · Hard Question Practice

Overview

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25 SAT-style challenge problems

Question 1

For the function

$$f(x) = a\sqrt{b-x} + \frac{12}{\sqrt{b-x}},$$

a and b are constants, and $f(4) = 0$. If $f(5) > 0$, which of the following could be a value for $a + b$?

- A. -20
- B. -12
- C. -8
- D. -4

Answer 1

Correct Answer: D

Question 2

$$\frac{16}{a} = \frac{16}{b} + \frac{16}{c}$$

Which of the following expresses a in terms of b and c ?

- A. $a = \frac{b+c}{bc}$
- B. $a = \frac{bc}{b+c}$
- C. $a = bc(b+c)$
- D. $a = 16bc(b+c)$

Answer 2

Correct Answer: B

Question 3

$$\frac{ax^3 + bx^2 + cx + d}{(x^2 - 9)(x + 2)} = \frac{5}{x + 3} + \frac{6x}{x - 3}$$

What is the value of $a + b + c$?

Answer 3

Final Answer: 72

Question 4

There are two lawn mowing companies. Company A charges a flat fee of \$80 for the first hour, then an hourly rate of \$60 for each additional hour. Company B charges no flat fee, but an hourly rate of \$75 per hour. After how many minutes will company B's total cost exceed company A's?

Answer 4

Final Answer: 80

Question 5

$$\frac{5}{6}x + ay = b$$

$$\frac{11}{10}x + cy = 2d$$

If the system of equations above has infinitely many solutions, what must be the value of $\frac{b}{d}$?

Answer 5

Final Answer: $\boxed{\frac{50}{33}}$

Question 6

a is 540% more than b and b is 0.02% of c . If a is $k\%$ of c , then what must k equal?

Answer 6

Final Answer: 0.128

Question 7

$$f(t) = a \cdot 2^{\frac{2t}{11}}$$

The given function $f(t)$ models the amount of money in an investment account t years after the account is opened, where a is a constant. How many months will it take for the initial investment to increase by 700%?

Answer 7

Final Answer: 198

Question 8

The function f is defined by

$$f(x) = k^{3kx} + b$$

for $x \geq 0$. k and b are positive constants and $k < 1$. Which of the following must be true?

- I. The maximum value of the function is displayed as a constant or coefficient of the function.
- II. $f(1) = f(0) \cdot k$
- III. $f(a) < b$ for some $a > 0$

- A. I only
- B. II and III only
- C. I and III only
- D. Neither I, II, nor III

Answer 8

Correct Answer: D

Question 9

$$15x^{18} + kx^9 + 24$$

The expression above has factors $ax^9 + b$ and $cx^9 + d$, where a , b , c , and d are all integer constants. What is the maximum value of k ?

Answer 9

Final Answer: 361

Question 10

$$(hx + k)(qx + z) = ax^2 + bx + 35$$

For the expression above, h , k , q , and z are all integer constants. Which of the following must be an integer?

- I. $\frac{35}{k}$
- II. b
- III. $-\frac{a}{q}$

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II, and III

Answer 10

Correct Answer: D

Question 11

$$f(x) = -3x^2 + bx + c$$

In the given quadratic function, b and c are constants. The graph of $y = f(x)$ in the xy -plane is a parabola that has a vertex at the point (h, k) , where h and k are constants. If $f(-4) = f(8)$, and the quadratic function has x -intercepts at $(r, 0)$ and $(s, 0)$, which of the following must be true?

- I. $r + s = 4$
 - II. $b < 0$
 - III. $c < k$
- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II, and III

Answer 11

Correct Answer: C

Question 12

A quadratic function models a rock's height, in feet, above the ground in terms of the time, in seconds, after it is thrown. The rock was thrown from an initial height of c feet above the ground and reached a maximum height of 200 feet above the ground 3 seconds after it was thrown. Then it hit the ground 8 seconds after being thrown. At what initial height c was the rock thrown?

Answer 12

Final Answer: 128

Question 13

$$2rx^2 - 5sx + 8r = 0$$

If the quadratic equation above has one real solution and $rs = 160$, what is $|r + s|$?

Answer 13

Final Answer: 26

Question 14

$$f(x) = -(x + a)^2(x - b)(x - c)$$

For the polynomial function above, a , b , and c are positive constants. Let $b < k < c$ for some constant k . Which of the following must be true?

- I. $k > 0$
 - II. $f(k) > 0$
 - III. $f(0) > 0$
-
- A. I and II only
 - B. II and III only
 - C. I and III only
 - D. I, II, and III

Answer 14

Correct Answer: A

Question 15

Three consecutive even integers are ordered from smallest to largest. 5 times the first integer is at least 9 more than the sum of the second and third integers. What is the minimum possible value for the second integer?

Answer 15

Final Answer: 8

Question 16

If a car accelerates at 108 kilometers per hour squared, what is its acceleration in meters per minute squared? (1 kilometer = 1000 meters)

Answer 16

Final Answer:

Question 17

In triangle $\triangle ABC$, the measure of $\angle B$ is 90° , AB is a leg with a length of 12, and point D lies on segment AC . Which of the following must be true?

- I. $BC = 12 \tan(\angle CAB)$
- II. $\cos(\angle ABD) - \sin(\angle DBC) = 0$
- III. $\angle ABD = \angle BDC$

- A. I and II only
- B. II and III only
- C. I and III only
- D. I, II, and III

Answer 17

Correct Answer: A

Question 18

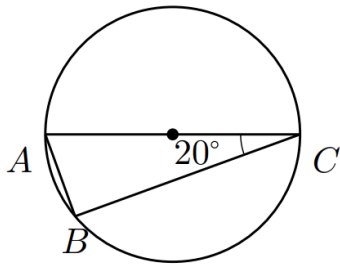
In triangle $\triangle ABC$, the measure of $\angle B$ is 90° , and BD is an altitude of the triangle. The length of BC is 10 and $\angle ABD = 60^\circ$. What is the length of AD ?

Answer 18

Final Answer: 15

Question 19

In the circle above, AC is a diameter and $\angle ACB = 20^\circ$. If the circumference of the circle is 90 inches, what is the length of arc BC ?



Answer 19

Final Answer: 35

Question 20

In the xy -plane, the graph of $x^2 + 6x + y^2 + 2y = 15$ is a circle. Line k has a slope of $\frac{3}{4}$ and is tangent to this circle at the point (a, b) , where b is negative. What is the value of b ?

Answer 20

Final Answer: -5

Question 21

Rectangular prism A and rectangular prism B are similar. The height of rectangular prism A is 4 inches and the height of rectangular prism B is 10 inches. If the volume of prism A is 200 cubic inches, then what is the volume of prism B ?

Answer 21

Final Answer: 3125

Question 22

Consider a square right pyramid with a volume of 400 cubic inches and a slant height of 13 inches. The pyramid's length, width, and height (in inches) are all integers. What is the surface area of the pyramid in square inches?

Answer 22

Final Answer: 360

Question 23

Data set A consists of 4 integers greater than or equal to 10 but less than or equal to 20. Data set B consists of the same four integers and adds one more integer, which must also be between 10 and 20. What is the largest possible difference between the mean of data set A and the mean of data set B ?

Answer 23

Final Answer: 2

Question 24

Data	Frequency
1	2
2	5
3	6
4	5
5	2

The frequency table above summarizes data set A . Data set B is created by multiplying all of the values in data set A by 2. Which of the following must be true?

- I. The difference between the mean and median of data set B is 0.
- II. The standard deviation of data set B is equal to the standard deviation of data set A .
- III. The range of data set B is double the range of data set A .

A. I and II only

B. II and III only

Answer 24

Correct Answer: C

Question 25

A sample of students was selected at random from all 11th graders at a school. Based on the sample, it is estimated that the proportion of 11th graders in at least one AP class is 60% with a margin of error of 5%. Which of the following conclusions are appropriate?

- I. It is not possible for just 40% of the 11th graders in the school to be in at least one AP class.
- II. It is plausible that the proportion of all students at the school in at least one AP class is between 45% and 55%.
- III. If the sample size is increased, the margin of error will drop below 5%.

- A. I and II only
- B. II and III only
- C. III only
- D. I, II, and III

Answer 25

Correct Answer: C

Thank You!

We appreciate your time and focus.

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